

Design Proposal: Mach-Zehnder Interferometers for Waveguide Group Index Extraction

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Abstract

This proposal covers three unbalanced Mach-Zehnder interferometers on 220 nm silicon-on-insulator, built from 500 nm wide strip waveguides for quasi-TE light at 1550 nm, with the aim of extracting the waveguide group index from the measured free spectral range.

1 Introduction

The objective is to build interferometer circuits from which the waveguide group index can be extracted, and later to compare the simulated value with the value measured on the fabricated chip. Silicon photonic interferometers are the basic building block of optical modulators, switches and wavelength filters, so an accurate group index is needed for almost every later design.

The design uses the default options of the course, listed in Table 1.

Table 1: Design choices.

Design choice	Value
Waveguide type	Strip waveguide, silicon core in silicon dioxide cladding
Waveguide height	220 nm (fixed by the process)
Waveguide width	500 nm
Polarization	Quasi-TE
Centre wavelength	1550 nm
Interferometer type	Mach-Zehnder (MZI)
Input splitter	Y-branch (<code>ebeam_y_1550</code>)
Output combiner	2×2 broadband directional coupler (<code>ebeam_bdc_te1550</code>), two output ports
Fibre coupling	TE grating couplers (<code>ebeam_gc_te1550</code>), 127 μm pitch
Bend radius	5 μm
Design variations	3 devices with path length differences of 9.776, 113.353 and 749.043 μm

The three path length differences span almost two orders of magnitude. This gives a wide range of free spectral ranges (about 59, 5.1 and 0.77 nm), so the trend of FSR against ΔL can be checked and the group index can be extracted with increasing precision.

2 Theory

2.1 Transfer function of the interferometer

The Y-branch splits the input field equally into two arms of length L_1 and L_2 . Each arm adds a phase and a loss set by the complex propagation constant:

$$E_{1,2} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_{1,2} - \frac{\alpha}{2} L_{1,2}}, \quad \beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (1)$$

The output combiner in this design is a 2×2 50/50 directional coupler, not a second Y-branch. An ideal coupler adds a 90° phase to the cross-coupled field, so the two output fields are

$$E_{o1} = \frac{1}{\sqrt{2}} (E_1 + iE_2), \quad E_{o2} = \frac{1}{\sqrt{2}} (iE_1 + E_2). \quad (2)$$

For negligible loss, and with $\Delta L = L_2 - L_1$, the intensity transfer functions of the two outputs are

$$T_{1,2}(\lambda) = \frac{I_{o1,2}}{I_i} = \frac{1}{2} [1 \pm \sin(\beta \Delta L)] = \frac{1}{2} \left[1 \pm \sin\left(\frac{2\pi n_{\text{eff}}(\lambda)}{\lambda} \Delta L\right) \right]. \quad (3)$$

The two outputs are complementary: $T_1 + T_2 = 1$, and a minimum on one port is a maximum on the other. Compared with the Y-branch combiner, which gives $\frac{1}{2}[1 + \cos(\beta \Delta L)]$, the fringes are shifted by a quarter period, but the period is the same. With loss included, the general form used for the simulations is

$$T_1(\lambda) = \frac{1}{4} \left| e^{-i\beta L_1 - \frac{\alpha}{2} L_1} + i e^{-i\beta L_2 - \frac{\alpha}{2} L_2} \right|^2. \quad (4)$$

2.2 Group index from the free spectral range

Two neighbouring minima (or maxima) of one output, at wavelengths $\lambda_1 < \lambda_2$, differ in phase by exactly 2π :

$$\beta(\lambda_1) \Delta L - \beta(\lambda_2) \Delta L = 2\pi. \quad (5)$$

For a small spacing $\text{FSR} = \lambda_2 - \lambda_1$, the left side is approximated by the first derivative of β :

$$-\frac{d\beta}{d\lambda} \text{FSR} \Delta L \approx 2\pi, \quad \frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda} \frac{dn_{\text{eff}}}{d\lambda} - \frac{2\pi n_{\text{eff}}}{\lambda^2} = -\frac{2\pi}{\lambda^2} \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right). \quad (6)$$

The term in brackets is the definition of the group index, which gives the two working equations of this project:

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}, \quad \text{FSR} = \frac{\lambda^2}{n_g \Delta L} \iff n_g = \frac{\lambda^2}{\text{FSR} \cdot \Delta L}. \quad (7)$$

In practice λ^2 is replaced by the product $\lambda_1 \lambda_2$ of the two fringe wavelengths, and the result is assigned to the midpoint wavelength.

3 Modelling and simulation

3.1 Simulation method

The waveguide was simulated with a full-vectorial finite-difference mode solver in Python (EMPy, VFDMODESolver [?]). The mode profiles in Fig. 1 were simulated in Lumerical MODE. The cross-section is a $500 \text{ nm} \times 220 \text{ nm}$ silicon core fully surrounded by silicon dioxide, on a 10 nm uniform mesh in a $3.0 \mu\text{m} \times 2.2 \mu\text{m}$ window. Material dispersion of Si and SiO_2 uses the Lorentz models from Chrostowski and Hochberg [1], which give $n = 3.4734$ and 1.4433 at 1550 nm .

Refining the mesh to 5 nm changes the effective index by 0.0013 , so the third decimal place of the effective index is the accuracy limit of this model. The values below agree with the published Lumerical results for this waveguide ($n_{\text{eff}} \approx 2.44$, $n_g \approx 4.18$) [1].

3.2 Mode profile

At 1550 nm the waveguide supports a fundamental quasi-TE mode ($n_{\text{eff}} = 2.446$, 97% of the transverse electric field energy in E_x) and a fundamental quasi-TM mode ($n_{\text{eff}} = 1.777$). A second TE-like mode appears at $n_{\text{eff}} = 1.496$, only 0.05 above the cladding index, so it is very weakly guided and is lost in the 5 μm bends. The design therefore operates as a single-mode TE circuit.

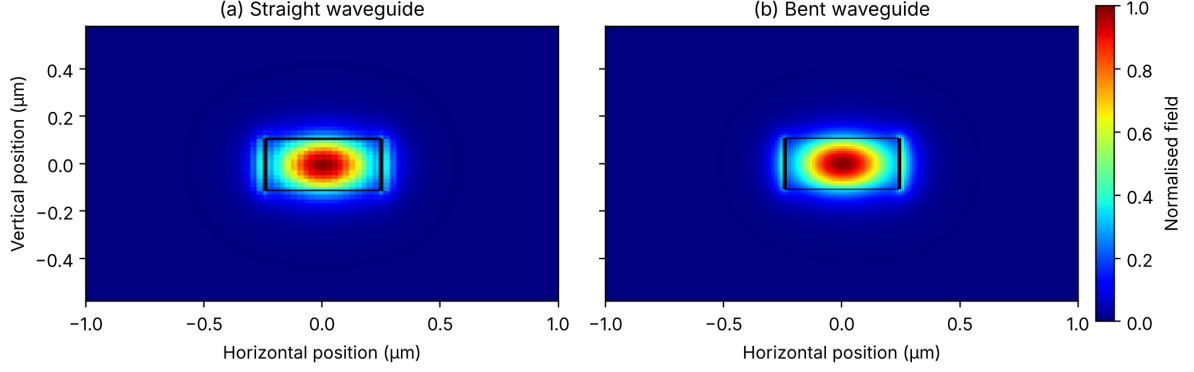


Figure 1: Fundamental quasi-TE mode of the 500 nm $\times$ 220 nm waveguide, simulated in Lumerical MODE (normalised): (a) straight waveguide, (b) bent waveguide. Both panels use the same scale, with the origin at the centre of the silicon core (black rectangle). In the bend the mode stays confined in the core, with its centre moved slightly towards the outside of the bend.

3.3 Effective index, group index and compact model

The wavelength was swept from 1500 to 1600 nm in 10 nm steps. The effective index falls almost linearly, from 2.5016 at 1500 nm to 2.3903 at 1600 nm. The group index is nearly flat: 4.167 at 1500 nm, 4.172 at 1550 nm and 4.177 at 1600 nm (Fig. 2).

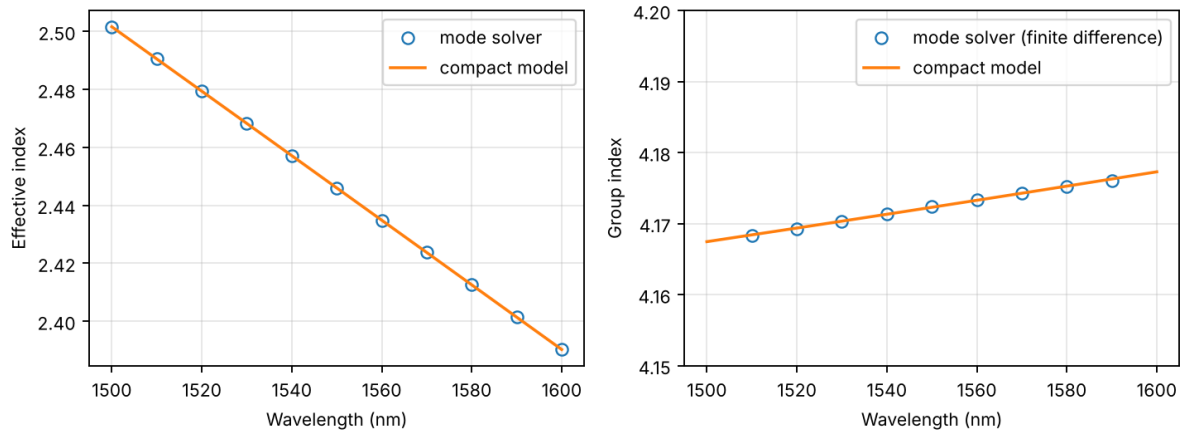


Figure 2: Effective index (left) and group index (right) of the quasi-TE mode. Circles are mode solver results; lines are the compact model.

A second-order polynomial around $\lambda_0 = 1.55 \mu\text{m}$ fits the effective index with a maximum residual of 1.3×10^{-6} (λ in μm):

$$n_{\text{eff}}(\lambda) = 2.4460 - 1.1137(\lambda - 1.55) - 0.0317(\lambda - 1.55)^2. \quad (8)$$

The group index of the compact model at 1550 nm is $n_g = 2.4460 + 1.55 \times 1.1137 = 4.172$.

3.4 Parameter variations and expected performance

All three devices share the same short arm and differ only in the long arm (Table 2). Lengths are the values stored in the layout and were confirmed from the drawn silicon area.

Table 2: Parameter variations and expected performance.

Device	L_1 (μm)	L_2 (μm)	ΔL (μm)	FSR at 1550 nm, $\lambda^2/(n_g\Delta L)$ (nm)	Minima per output, 1500 to 1600 nm	Extra loss in long arm at 3 dB/cm (dB)
MZI1	19.572	29.348	9.776	58.90	2	0.003
MZI2	19.572	132.925	113.353	5.08	20	0.03
MZI3	19.572	768.615	749.043	0.769	130	0.22

The course design check asks for an FSR smaller than the measurement bandwidth of about 50 nm. MZI2 and MZI3 pass this check with a large margin. MZI1 does not: its FSR of about 59 nm is larger than 50 nm, so at most one full fringe is visible per output. See Section ?? for the recommended change.

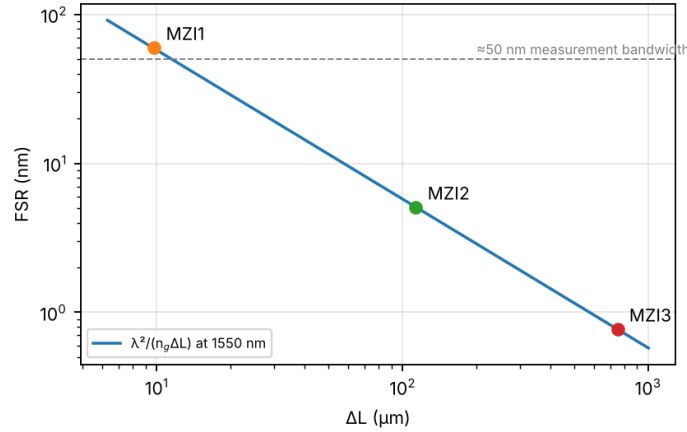


Figure 3: FSR against ΔL at 1550 nm (line) and the three devices (points, from the analytic model spectra). The dashed line is the 50 nm measurement bandwidth.

3.5 Circuit transmission spectra

The three circuits were simulated in Lumerical INTERCONNECT from the layout netlist, using the compact models of the EBeam PDK for the grating couplers, the Y-branch, the broadband directional coupler and the waveguides. The laser sweep is 1499 to 1601 nm. Figure 4 shows the two output ports of each circuit in dB.

The FSR values in Table 3 were read from the plots, so they carry a reading uncertainty of about 0.5 %. They are compared with an analytic model that uses the compact waveguide model of Eq. (8) in the transfer function of Eq. (4), with ideal 50/50 splitters.

MZI2 and MZI3 agree with the analytic model and with the simulated waveguide group index of 4.172 within the reading uncertainty. MZI1 does not: its FSR is about 10 % larger than $\lambda^2/(n_g\Delta L)$. The real splitter and coupler add a small wavelength-dependent phase difference, and with a path difference of only 9.776 μm this phase is a visible part of the total. For the longer devices the same phase is negligible.

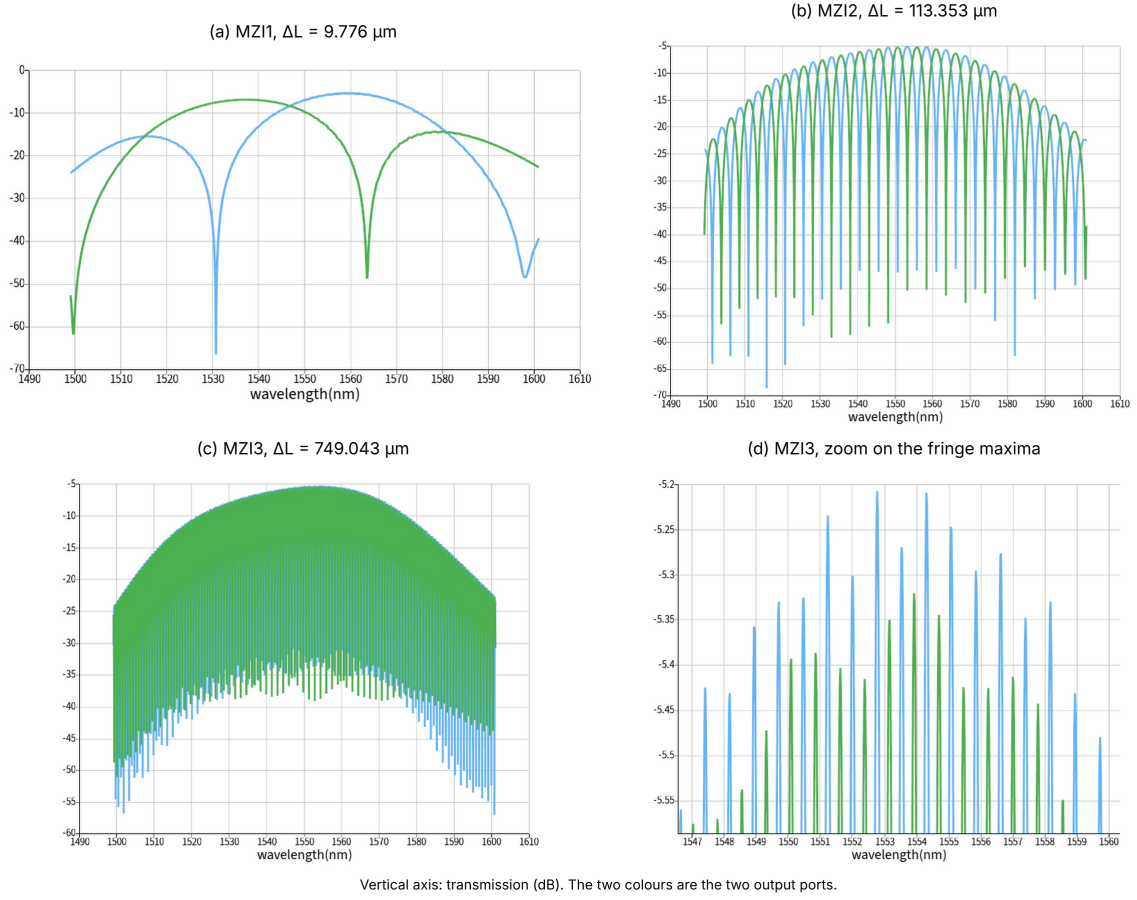


Figure 4: INTERCONNECT transmission spectra, both output ports: (a) MZI1, with only two minima per output inside the sweep; (b) MZI2, with 20 minima per output; (c) MZI3 over the full sweep, where the fringes are too dense to resolve; (d) zoom on the fringe maxima of MZI3 between 1547 and 1560 nm, where neighbouring maxima of one output are about 0.77 nm apart.

Table 3: FSR from the analytic model and from INTERCONNECT.

Device	FSR, analytic model (nm)	FSR, INTERCONNECT (nm)	How it was read	n_g from INTERCONNECT
MZI1	59.7	about 64 to 67	Minima near 1530.7 and 1598 nm (one output), 1499.7 and 1563.5 nm (other output)	about 3.72
MZI2	5.07	about 5.08	19 periods between minima near 1501.5 and 1598 nm	about 4.17
MZI3	0.769	about 0.769	16 periods between maxima near 1547.4 and 1559.7 nm	about 4.19

3.6 Test of the group index equation

The equation $n_g = \lambda_1 \lambda_2 / (\text{FSR} \cdot \Delta L)$ was applied to the two minima of output 1 closest to 1550 nm in each spectrum of the analytic model with ideal splitters, and compared with the group index of the compact model at the midpoint wavelength (Table 4).

Table 4: Test of the group index extraction on the analytic model spectra.

Device	λ_1 (nm)	λ_2 (nm)	FSR (nm)	Extracted n_g	Model n_g at midpoint	Difference
MZI1	1531.220	1590.930	59.710	4.1733	4.1734	0.002 %
MZI2	1545.594	1550.662	5.068	4.1720	4.1721	0.002 %
MZI3	1549.463	1550.232	0.769	4.1701	4.1723	0.05 %

The equation recovers the model value in all three cases. The larger difference for MZI3 comes from the 1 pm wavelength grid of the simulation: one grid step is 0.13 % of a 0.769 nm FSR. The same effect will be stronger in the experiment, where the laser step is typically 10 pm.

4 Layout

4.1 Overview

The layout file `EBeam_hesaamz.gds` was drawn in KLayout 0.30.12 with SiEPIC-Tools 0.5.31 and the EBeam PDK 0.4.53 [5]. The top cell `EBeam_hesaamz_v1` holds three device cells, MZI1, MZI2 and MZI3, placed side by side (Fig. 5).

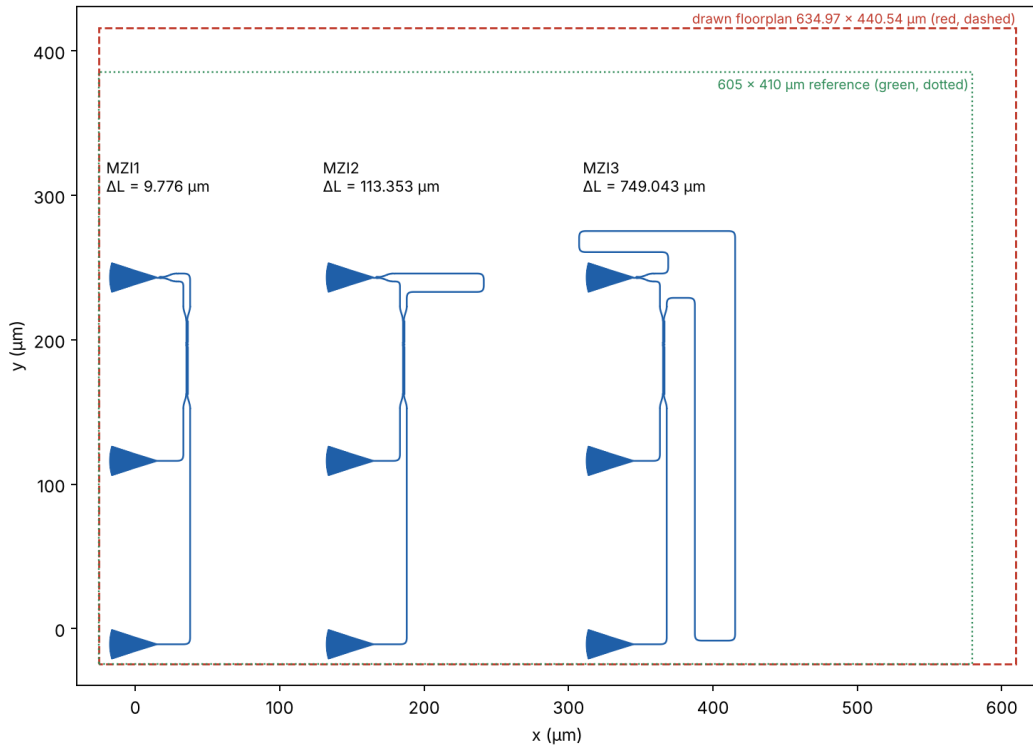


Figure 5: Silicon layer of the full layout, read from the GDS file. The red dashed box is the drawn floorplan; the green dotted box is a $605 \mu\text{m} \times 410 \mu\text{m}$ reference with the same lower-left corner.

4.2 Circuit of each device

Every device has the same structure, and only the long arm changes:

1. Three TE grating couplers [2] in a vertical column at $127\text{ }\mu\text{m}$ pitch, facing the same direction. The top coupler is the input and carries the `opt_in` label.
2. A Y-branch [3] directly at the input coupler splits the light into the two arms.
3. The short arm ($19.572\text{ }\mu\text{m}$) and the long arm end at the two inputs of the broadband directional coupler [4].
4. The two coupler outputs go to the middle grating coupler ($53.328\text{ }\mu\text{m}$ of waveguide) and to the bottom grating coupler ($185.028\text{ }\mu\text{m}$ of waveguide).

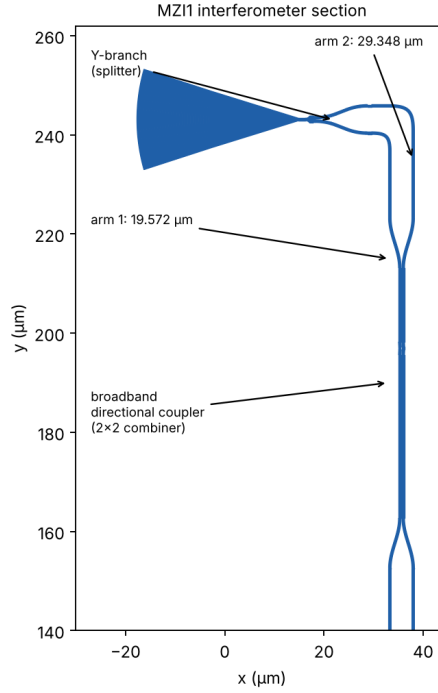


Figure 6: Interferometer section of MZI1: Y-branch splitter, the two arms, and the 2×2 broadband directional coupler used as combiner.

References

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*, Cambridge University Press, 2015.
- [2] Y. Wang *et al.*, “Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits,” *Optics Express* **22**, 20652 (2014). doi:10.1364/OE.22.020652
- [3] Y. Zhang *et al.*, “A compact and low loss Y-junction for submicron silicon waveguide,” *Optics Express* **21**, 1310 (2013).
- [4] Z. Lu *et al.*, “Broadband silicon photonic directional coupler using asymmetric-waveguide based phase control,” *Optics Express* **23**, 3795 (2015).
- [5] KLayout with SiEPIC-Tools and the SiEPIC EBeam PDK, used for the layout and its verification.